

Intro to Algorithms:

An **algorithm** is a finite, well defined sequence of computation steps that takes some value or ~~value~~ set of values, as input & produces some value or set of values as output.

3 core components:

1) Input:

2) Instructions / logical steps.

3) Output.

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Correctness of an Algorithm:

An algorithm is considered correct, if for every valid input, it halts & produces the expected, accurate Output.

Incorrect if:

- wrong / incorrect output
- enter an infinite loop.

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Why study Algorithms?

- 1) Performance & Scalability
- 2) Resource Efficiency
- 3) Problem-Solving Framework
- 4) Foundation for computer science.

Writing Pseudocode

In this course we will be machine / language independent so we will write pseudocode.

Basic ~~set~~ structure:

- 1) Always start with **"BEGIN"** & end with **"END"**
- 2) Declare variables before using using standard data types:
 - "NUMBER"** (for int & floats).
 - "STRING"** (text)
 - "BOOLEAN"** (true/false).
- 3) Input/Output use key word:
 - INPUT** variable
 - OUTPUT** text or variable
- 4) Assignment: use **" = "** to store values into variable.

5) Conditionals:

```
1)      IF      condition      THEN
          -----
          .... body ....
          -----

      END IF
```

```
2)      IF      condition      THEN
          -----
          .... body ....
          -----

      ELSE IF      condition      THEN
          -----
          .... body ....
          -----

      ELSE
          -----
          .... body ....
          -----

      END IF
```

* Else doesn't have a THEN..

6) WHILE loop:

WHILE condition

---body---

ENDWHILE

7) FOR loop:

we only used this when we know how many time a block of code should repeat.

~~FOR~~

a)

NUMBER count

FOR count = 1 TO ~~count~~ 5

---body---

END FOR.

By default for is considered with one step & ascending.

B)

NUMBER i

FOR $i=1$ TO 6 STEP 2

-- body

END FOR.

C) (decending).

NUMBER i

FOR $i=10$ TO 1 STEP -1

-- body.

END FOR

8) General points:

- In ~~code~~ pseudocode, **indentation matters**.
without it, code becomes impossible to read.

Examples:

Q INPUT Age & check eligibility for driving.

BEGIN

NUMBER Age;

OUTPUT "What is your Age? "

INPUT Age

IF Age $\geq$ 18 THEN

OUTPUT "You can drive"

ELSE

OUTPUT "You cannot drive"

END IF

END.

(Q2) Read 10 numbers & find sum of even Numbers.

BEGIN

NUMBER i, sum, current Number

sum = 0

FOR i = 1 **TO** 10 **STEP** 1

OUTPUT "Enter Number " + i + " : "

INPUT current Number

IF current Number % 2 == 0 **THEN**

sum += current Number

END IF

END FOR

OUTPUT "Sum of Even: " + sum

END

Algorithm Characteristics

The necessary features of an algo:

1) Definiteness:

- The steps of an algo must be precisely defined.

2) Effectiveness:

- Individual steps are all do-able

3) Finiteness:

It won't take forever to produce the desired output for any input in the specified domain.

4) Output:

Information or data that goes out. There should be at least one output that corresponds to the input.